

YUSEF SYED

Toronto, ON | U.S. and Canadian citizen | Open to relocation
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EDUCATION

University of Toronto

Expected May 2030

Bachelor of Science – Mathematical & Physical Sciences (first year)

Toronto, ON

Intended double major: Mathematics and Computer Science.

Coursework: programming, calculus with proofs, linear algebra; prior Data Structures (dual enrollment).

PROJECTS

Providence | *Swift, SwiftUI, watchOS* | [Devpost](#)

Sep 2026

1st Place, Aramco Challenge – Hack the North 2026

- Integrated a native Apple Watch client with the team's voice-controlled computer assistant, supporting hold-to-talk capture, paired-Mac transcription, and spoken-result playback.
- Implemented routing to the selected Mac and chat, preserving request IDs across retries and rejecting stale approvals or results bound to another request.

Aesthetics AI | *TypeScript, React Native, PostgreSQL* | [App Store](#)

May 2026 – Present

- Released an iOS fitness app with AI photo reports, workout tracking, and subscriptions on the App Store.
- Hardened workout saving and atomic sync, with **47 local behavioral tests** covering recovery and account isolation.

Agent Eval Mutation Lab | *Python, Inspect, Docker, SQLite* | [GitHub](#)

Aug 2026 – Present

- Built a Python evaluation engine to test whether proposed tool actions match recorded effects across **104 synthetic configurations**, using transactional SQLite checkpoints and deterministic scheduling and exports.
- Tested tool execution across **13 controlled Docker fault fixtures**, using independent SQLite checks to distinguish denied requests, committed side effects, rollbacks, and retries rather than relying on response text alone.

Tiraz Garment-Category Completion | *Python, PyTorch* | [GitHub](#)

Jun 2026 – Present

- Trained three seeded set encoders on 27,683 annotation groups for category completion; improved top-1 accuracy by **5.8–6.3 percentage points** over a 52.7% co-occurrence baseline on 962 held-out groups, using separate tuning and calibration splits.

OPEN SOURCE CONTRIBUTIONS

Open-Source Contributor | Merged PRs

Aug 2026 – Present

- Contributed **12 merged upstream PRs** to six projects, including Apache Arrow, Microsoft PyRIT, and Kornia.
- Fixed [Apache Arrow](#) slicing to accept Arrow integer scalars across four Python APIs; added regression tests for integer types, invalid inputs, and overflow behavior.
- Fixed [Microsoft Agent Lightning](#) to preserve subprocess timeout failures across status-report retries; added regression tests for reporting failures, exit-code races, and shutdown reconciliation.

SUPERVISED RESEARCH

Conformal Prediction – Beginning Supervised Work

Sep 2026 – Present

Supervisor: Prof. Pankaj Bhagwat, Department of Statistics, University of Manitoba.

Studying conformal prediction and its statistical assumptions; computational experiments planned under supervision.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Swift

Frameworks & data: React Native, SwiftUI, PyTorch, Inspect, PostgreSQL, SQLite

Tools & testing: Git, GitHub Actions, Docker, pytest, Xcode

LEADERSHIP

Feeding the Valley | Vice President

2025 – 2026

Helped coordinate 60+ volunteers in an initiative that raised \$20,000 and prepared 1,700 meals for local food banks.

FIRST Robotics Team 834 | Fundraising Lead

2022 – 2026

Supported fundraising and outreach totaling \$5,000+ raised and 2,000+ prospects contacted.